Muhammad Haseeb

haseeb099m@gmail.com | github.com/ha405 | linkedin.com/in/muhammad-haseeb

RESEARCH INTERESTS

AI Safety; Mechanistic Interpretability; Emergent Misalignment; Domain Generalization; Efficient ML.

EDUCATION

Lahore University of Management Sciences (LUMS) Sep 2022 – May 2026
B.S. Computer Science
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, AI on Edge Devices, Advanced Topics in ML, LLM Systems, Linear Algebra, Probability.

PUBLICATIONS

- **Do Emergent Misalignment Models Share a Circuit? (*arXiv preprint; research in progress*).**
  Muhammad Haseeb, Muhammad Tahir.
- **Mechanistic Evidence for Preserved-but-Misaligned Representations in Non-IID FedAvg (*ICML 2026 Mechanistic Interpretability Workshop*).**
  Muhammad Haseeb, Salaar Masood, Muhammad Abdullah Sohail, Mohammad Fatim Shoaib, Muhammad Tahir.
- **BaCP: Backbone Contrastive Pruning for Preserving Representations in Extremely Sparse Neural Networks (*Preprint*).**
  Mohammad Haroon Khawaja, **Muhammad Haseeb**, Mohammad Fatim Shoaib, Muhammad Tahir.

RESEARCH EXPERIENCE

ML Research Assistant Jan 2025 – May 2026
Centre for Urban Informatics, Technology and Policy (CITY), LUMS
Advisors: Dr. Muhammad Tahir, Dr. Zubair Khalid

- **Emergent Misalignment - AI Safety:**
  - Identified three attention-head locations shared across medical, finance, and sports fine-tunes, extending evidence of common structure beyond representations.
  - Reduced generated misalignment by **16–38 percentage points** with targeted interventions, without degrading judged coherence.
- **Mechanistic Analysis of Federated Learning:** (*Accepted at ICML 2026 Mechanistic Interpretability Workshop*)
  - Studied FedAvg through the lens of Mechanistic Interpretability to analyze performance loss under Non-IID conditions using class-specific circuit discovery, Universal Sparse Autoencoders (USAEs), and linear probes.
  - Compared class-specific circuits and observed drift and interference patterns during non-IID aggregation, while IID circuits behaved in a perfect constructive manner.
  - Demonstrated via USAE concept analysis that useful latent features, similar to the IID feature basis, remain present and recoverable even when global accuracy degrades under label skew.
- **Backbone Contrastive Pruning (BaCP):**
  - Contributed to a generalized pruning framework to merge standard pruning criteria with contrastive learning to prevent representational collapse.
  - Designed a multi-objective loss function that aligns sparse embeddings with pretrained, fine-tuned, and historical snapshot references to maintain feature consistency with pretrained references to prevent representational collapse.
  - Maintained accuracy comparable to dense baselines at up to **99% sparsity**, outperforming standard unstructured pruning approaches across ViTs/CNNs/Language Models.
- **Domain Generalization & Interpretability:**
  - Applied circuit extraction techniques to isolate task-specific subnetworks, analyzing whether out-of-distribution failures are caused by shortcut learning or weight interference.

- Optimized visual queries in Vision Transformers using reinforcement learning (Group Relative Query Optimization), achieving a **3% accuracy gain** on the PACS dataset compared to empirical risk minimization.

– **Quantization of Diffusion Models:** Proposed Log-SNR conditioned temporal dynamic quantization (**55% lower LPIPS** on PixArt-α) and implemented Hessian-based GPTQ/QroNos for **4-bit weight compression**.

AI/SWE Research Intern May 2025 – Aug 2025
University of Illinois Urbana-Champaign (UIUC)

– Investigated LLM-based heuristics for automated `#ifdef` guard insertion in C codebases and explored large language models for software debloating and code dependency analysis.

– Contributed to the development of a VS Code extension supporting a C-to-Rust translation pipeline.

ML Research Assistant Jan 2025 – Aug 2025
Computer Vision & Graphics Lab, LUMS

– **KL Aware Quantization (KLAWQ):** Proposed an augmented GPTQ framework integrating KL divergence and second-order curvature to optimize a combined MSE+KL objective, reducing LLM perplexity by **30%** compared to GPTQ baselines.

– **Single-Image Camera Calibration:** Developed a pipeline to train a Multi-Scale Deformable Transformer by combining loss with geometric constraints to recover full camera projection matrices from a single image.

INDUSTRY EXPERIENCE

Founding Product Engineer June 2026 – Present
NEXA (Remote)

– Architecting and building highly scalable, distributed backend systems from the ground up.

– Integrating machine learning models and optimizing distributed serving pipelines for fast, efficient inference.

OSS Engineer Dec 2025 – Feb 2026
DatacurveAI (YC S24)

– Solved bugs and added new features across various open-source machine learning repositories to test and train LLMs on working with large codebases.

Machine Learning Engineer Jul 2025 – Oct 2025
Innova Tech

– Automated data annotation pipeline, refined training configurations, and fixed model architectures getting a **4% accuracy gain**. Also, implemented inference optimization with TensorRT for speedup and various quantization techniques to reduce memory footprint by **40%** on edge devices.

TEACHING EXPERIENCE

Teaching Fellow, MSAI Aug 2026 – Present
AI 630: Responsible AI Engineering, LUMS

– Developed and graded quizzes for AI 630: Responsible AI Engineering.

– Managed class logistics and coordinated course projects.

Teaching Assistant Aug 2025 – Dec 2025
CS436: Computer Vision, LUMS

– Designed and supervised an assignment on transfer learning and multithreaded C++ object detection. Mentored 40 student groups in building a virtual tour application using Structure from Motion.

TECHNICAL SKILLS

Python, C, C++, Rust, SQL; PyTorch, Transformers, ONNX, TensorRT, diffusers, adapters.